Algebra 1 – Module 5 Practice Test •
WORKED KEY

Full Name: ______________________________

Period: _________

This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

1. Write the equation, in slope-intercept form, of the line with a y-intercept of -4 and a slope of 5.

Answer: $y = 5x - 4$

2. Write the equation, in slope-intercept form, of the line through $(3, 4)$ and $(5, 14)$.

Answer: $m = 10/2 = 5$; $4 = 5(3) + b \to b = -11 \to y = 5x - 11$

3. Write the equation, in point-slope form, of the line through $(-4, -9)$, with a slope of 5/2.

Answer: $y + 9 = (5/2)(x + 4)$

4. Write the equation, in point-slope form, of the line through $(5, 4)$ and $(9, 16)$.

Answer: $m = 12/4 = 3 \to y - 4 = 3(x - 5)$

5. Rewrite $y + 5 = 4(x - 3)$ in slope-intercept form.

Answer: $y + 5 = 4x - 12 \to y = 4x - 17$

6. Rewrite $y = -(5/6)x + 4$ in standard form (integer coefficients).

Answer: multiply by 6: $6y = -5x + 24 \to 5x + 6y = 24$

7. Rewrite $14x - 7y = 21$ in slope-intercept form.

Answer: $-7y = -14x + 21 \to y = 2x - 3$

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8. Write the equation, in slope-intercept form, of the line that passes through (−2, 5) and is PARALLEL to $y = -4x + 1$.

Answer: same slope −4: $5 = -4(-2) + b \rightarrow b = -3 \rightarrow y = -4x - 3$

9. Write the equation, in slope-intercept form, of the line that passes through (3, −2) and is PERPENDICULAR to $y = (3/4)x + 6$.

Answer: slope −4/3: $-2 = -(4/3)(3) + b \rightarrow b = 2 \rightarrow y = -(4/3)x + 2$

10. Classify each pair of lines as parallel, perpendicular, or neither. Justify with slopes.

a. $y = -(3/7)x + 5$ and $y = -(3/7)x - 2$

Answer: parallel — equal slopes, different intercepts

b. $y = (1/6)x - 4$ and $y = -6x + 1$

Answer: perpendicular — slopes 1/6 and −6 are opposite reciprocals

11. Error Analysis — Rex says a line perpendicular to $y = -(3/8)x + 5$ has slope −8/3. Identify his mistake and give the correct slope.

Answer: he took the reciprocal but KEPT the negative sign — perpendicular slopes are OPPOSITE reciprocals; correct: 8/3

12. Marisol rents a kayak for an \$8 launch fee plus \$7 per hour. Write an equation for the total cost C after h hours, then find the cost after 5 hours.

Answer: $C = 7h + 8$; $C = 7(5) + 8 = \$43$

13. Write the equation, in STANDARD form, of the line with x-intercept (5, 0) and y-intercept (0, −4).

Answer: $m = 4/5,\ b = -4 \rightarrow y = (4/5)x - 4 \rightarrow 4x - 5y = 20$

Cumulative Review — ACT Style. Circle the best answer.

14. What is the slope of the line through (1, −3) and (3, 11)?

Ⓐ 1/7
Ⓑ **7**
Ⓒ 14
Ⓓ 28

Answer: 14/2 = 7

15. Let $f(x) = 6x - 5$. What is $f(4)$?

Ⓔ 14
Ⓕ **19**
Ⓖ 24
Ⓗ 29

Answer: 24 − 5 = 19

16. Solve: $3(x - 4) + 2x = 28$

Ⓐ 4
Ⓑ 32/5
Ⓒ **8**
Ⓓ 40

Answer: $5x - 12 = 28 \rightarrow x = 8$